IMPLEMENTATION OF DECISION MAKING AND KNOWLEDGE REPRESENTATION

Prolog Code:

```
% Rule to find the minimum of two numbers
minimum(X, Y, X) :- X =< Y.   % If X is less than or equal to Y, X is the minimum.
minimum(X, Y, Y) :- X > Y.    % If X is greater than Y, Y is the minimum.

% Rule to find the maximum of two numbers
maximum(X, Y, X) :- X >= Y.   % If X is greater than or equal to Y, X is the maximum.
maximum(X, Y, Y) :- X < Y.    % If X is less than Y, Y is the maximum.
```

Example Queries and Output:

```
1. To find the minimum of two numbers:
?- minimum(5, 10, Min).
Output:
Min = 5.

2. To find the maximum of two numbers:
?- maximum(5, 10, Max).
Output:
Max = 10.

3. Minimum and maximum together:
?- minimum(8, 3, Min), maximum(8, 3, Max).
Output:
Min = 3, Max = 8.
```

Prolog Code:

```
% Given facts
likes(mary, food).
likes(mary, wine).
likes(john, wine).
likes(john, mary).

% Rules based on the conditions:
likes(john, X) :- likes(mary, X).   % John likes anything that Mary likes
likes(john, Y) :- likes(Y, wine).   % John likes anyone who likes wine
likes(john, Y) :- likes(Y, Y).      % John likes anyone who likes themselves

% Sample queries:
% Query 1: Does John like food?
% ?- likes(john, food).
% Query 2: Does John like wine?
% ?- likes(john, wine).
% Query 3: Does John like food if Mary likes food?
% ?- likes(john, food).
% Query 4: Who does John like?
% ?- likes(john, Y).
```

Output:

```
Query: ?- likes(john, food).
yes

Query: ?- likes(john, wine).
yes

Query: ?- likes(john, food).
yes

Query: ?- likes(john, Y).
Y = mary ;
Y = john ;
Y = wine ;
```